Physics of the Atmosphere

Student sheet

Name: _____ **Date:** _____

Safety: On the stairs, walk, do not run; hold the box with both hands and read the screen only standing on a landing.

1. What I see on the screen

The screen shows **Temperature**, **Humidity** and **Pressure** (hPa, two decimal places) and, computed from them, **Dew point** and **Heat Index** (the felt temperature). **Heat Index** needs above 27 °C **and** above 40% humidity. If either is missing, the app shows the air temperature itself there — **do not record it as a result**.

2. What I do on my own

First I predict. Climbing the stairs, pressure will: ☐ increase ☐ decrease ☐ stay the same, by _____ hPa per storey.

because _____

_____

Building: _____ **One storey:** _____ m **Forecast QNH:** _____ hPa

Baseline measurement. T = _____ °C RH = _____ % P = _____ hPa

$z = 44330\,[1 - (P/P_0)^{0.1903}]$: with $P_0 = 1013.25$: _____ m; with P_0 = QNH: _____ m true: _____ m

$\gamma = \ln \frac{\mathrm{RH}}{100} + \frac{17.625\,T}{243.04+T}$ = _____ $T_d = \frac{243.04\,\gamma}{17.625-\gamma}$ = _____ °C $T - T_d$ = _____ °C

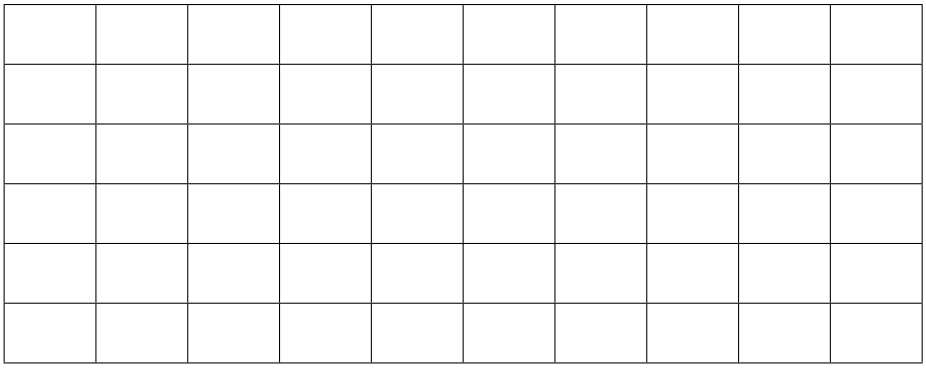
The traverse. 2–3 min per landing. $\ln P$ **to 5 decimal places:** 1008.40 ⇒ 6.91612.

Floor	z **(m)**	T **(°C)**	P **(hPa)**	$\ln P$ **(5 d.p.)**
Ground floor (mean)	0			
Floor 1				
Floor 2				
Top floor (mean)				
Ground floor (return)	0			

The check. Three readings at the ground floor and three at the top, 30 s apart; enter the mean.
Ground floor: _____ _____ _____ spread (max − min) = _____ hPa
Top floor: _____ _____ _____ spread = _____ hPa
Difference between levels = _____ hPa. A larger spread means **not detectable**, not that it does not exist: ☐ detectable ☐ not detectable
The return to the ground floor is not optional: a different reading means the weather changed and the gradient cannot be trusted. Difference _____ hPa. Trust it? ☐ YES ☐ NO

The graph. $\ln P$ vertical, z horizontal. **The vertical axis does not start at zero.**

top: _____



bottom: _____ Height z above the ground floor (m), from 0 to _____ — ten divisions

Gradient, from two points *on the line*: z_1 = ____ m, $\ln P_1$ = _____ z_2 = ____ m, $\ln P_2$ = _____
$m = (\ln P_2 - \ln P_1)/(z_2 - z_1)$ = _____ m^{-1} $H = -1/m$ = _____ m against 8400 m: _____ %

3. The conclusion I write

I predicted that, and the data show that _____

_____

The difference is explained by _____

_____

The gradient of my line gives the scale height ($H = -1/m$), so _____

_____

Exit ticket

One thing that surprised you today, and one thing you would do differently if you repeated the measurement.

_____